

Advanced OpenMP

Overview



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Partners



THE UNIVERSITY
of EDINBURGH

CRAY[®]
a Hewlett Packard Enterprise company





- UK National Supercomputer Service
 - managed by UKRI/EP SRC
 - to be housed, operated and supported by EPCC
 - hardware Supplied by Cray
- Training provided by the ARCHER2 Computational Science and Engineering (CSE) support team
 - 60 days per year at various locations
 - free to all academics



ARCHER2



- Full system has 5848 nodes
 - 128 cores per node, 748,544 cores
 - 256 GB per node (512 for some large memory nodes)
 - Cray Slingshot interconnect
 - 4 Lustre filesystems (14PB)
 - 1 Burst buffer filesystem (1PB)



Key ARCHER2 Resources

- Upcoming courses
 - <http://www.archer2.ac.uk/training/>
- Material from past courses
 - <https://www.archer2.ac.uk/training/materials/>
- Virtual tutorials (online)
 - <https://www.archer2.ac.uk/training/online/>
- Documentation
 - <https://www.archer2.ac.uk/documentation/>



Who am I?



Mark Bull m.bull@epcc.ed.ac.uk

- Senior research fellow at EPCC
- Lecture on EPCC's MSc in HPC
- EPCC's representative on the OpenMP ARB



Other Resources

- Please fill in the feedback form!
 - <https://www.archer2.ac.uk/training/feedback/>
- General enquiries about ARCHER go to the helpdesk
 - support@archer2.ac.uk
- EPCC runs one-year taught postgraduate masters courses
 - ***MSc in HPC*** and ***MSc in HPC with Data Science***
 - awarded by the University of Edinburgh since 2001
 - scholarships available
 - <http://www.epcc.ed.ac.uk/msc/>

Access to ARCHER2



- Your ARCHER2 `ta189` project accounts
 - Small amount of budget
 - `ta189` project accounts allow us to use a reservation to access dedicated compute nodes and get our jobs to run more quickly
 - Other ARCHER2 accounts can be used – use the short queue for quicker job turnaround
- Accounts will be closed two weeks after access ends
 - all files etc. will be deleted
 - take copies of all your work beforehand!
- Course materials (slides, exercises etc) available from ARCHER2 website
 - archived on ARCHER2 web pages for future reference



Code of Conduct



<https://www.archer2.ac.uk/training/code-of-conduct/>

- We expect all course trainers and attendees to:
 - Use welcoming and inclusive language
 - Be respectful of different viewpoints and experiences
 - Gracefully accept constructive criticism
 - Focus on what is best for the community
 - Show courtesy and respect towards other community members
- See web page for full details and incident reporting form



Timetable



- Tues 18th (10:00 - 12:00) Tasks, Nested parallelism, Memory model
- Thur 20th (10:00 - 12:00) OpenMP tips, tricks and pitfalls, Performance issues
- Tues 25th (10:00 - 12:00) OpenMP + MPI
- Thur 27th (10:00 - 12:00) Newer features in OpenMP

Lecture notes etc.



Go to the ARCHER2 site at <https://www.archer2.ac.uk/> and follow the links to this course under the Training tab



Practical exercises source code



To download the source code for the practical exercises, make sure you are in your **work** directory on ARCHER2, then use the following command :

```
cp /work/z19/shared/advomp.tar .  
tar xvf advomp.tar
```



I hope you enjoy the course



- ... and *please ask questions!*

